CAPSTONE PROJECT

Master in Business Analytics & Data Science Part-Time MBDS-PT FEB-2025 F-1

Area Functional Group - Program Direction

Number of sessions: 30

Term: 3rd online term

Category: regular

Language: English

Professor: **FEDERICO CASTANEDO SOTELA**

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Dr. Federico Castanedo currently works as Director of Applied AI at Inception, the research lab of G42. In his current role he leads the development of cutting- edge AI applications and models.
Prior to his role at Inception, he was Director of Data Science at DataRobot and the Head of Data Science at Vodafone. He also co-founded WiseAthena and served as its Chief Data Scientist.
Dr. Castanedo has a distinguished academic background, including his role as the former Academic Director of the Bachelor in Computer Science and Artificial Intelligence at IE University, where he continues to serve as an Advisor and Adjunct Professor.
He holds a Ph.D. in Computer Science and Artificial Intelligence from Universidad Carlos III de Madrid and conducted research as visiting scholar at Stanford University. His work includes several O'Reilly e-books and research papers that have received over 1,500.

Office Hours

Office hours will be on request. Please contact at:

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SUBJECT DESCRIPTION

The Capstone Project, also known as the Corporate Project, aims to allow students to showcase their mastery in applying the tools, knowledge, and skills acquired during the MBD program.

This project represents the final, comprehensive experience of their master's program. It requires students to develop, present, and defend an original work as a prerequisite for graduation.

The project offers an opportunity to address a real-world problem or challenge presented by an external company. This endeavor coincides with the final weeks of the MBD's Electives Period. Participation in the Impact Project entails accepting a business challenge from an external partner organization, conducting thorough analysis, formulating recommendations based on the provided data, and engaging in regular interactions with the IE Academic Tutor.

Students will collaborate in groups of **five to six** for the Impact Project. While part of the assessment is based on collaborative efforts, each team member will also receive an individual evaluation, reflecting their personal contributions and performance.

LEARNING OBJECTIVES

- Collaborate effectively within teams.
- Communicate and defend innovation proposals before audiences.
- Apply innovation methodologies (e.g., Design Thinking, Lean Startup, etc.) effectively to identify and validate problems and develop feasible solutions.

The student will have the opportunity to demonstrate their proficiency in tackling real-world business challenges through data-driven analytics. Partner companies will supply data sets, enabling students to collaborate and apply their skills. The capstone project serves as a platform for students to exhibit the knowledge and skills honed during the program by:

- Precisely **identifying** and **analyzing** the business challenge or problem, followed by proposing a solution grounded in data analytics.
- **Choosing** the most suitable data science model and methodology to effectively address the problem while considering the trade-offs among various solutions.
- **Creating** a sustainable architecture that supports the selected model.
- **Developing** a proof of concept utilizing real-world data and employing the tools and techniques learned throughout the program.

- Effectively **presenting** their proposal and findings to a panel with clarity and precision.

- **Acknowledging** and **addressing** any ethical or legal implications associated with their proposed solution.

The primary objective of this capstone project is to devise a data-driven solution for a business issue presented by a partner company, thereby transforming data into tangible business value

TEACHING METHODOLOGY

Participants will tackle a real-world business challenge set forth by a partner company, employing the skills and tools acquired during the program.

The company will supply a dataset and define their specific business objective. The ultimate goal is to develop a solution that will be presented to a tribunal comprising faculty members and experts from the company.

The Capstone Project is valued at 6 ECTS and unfolds through a blend of master classes, collaborative group work, and sessions with the academic tutor. These sessions are strategically designed to facilitate steady progress, offering support and guidance from the Academic Tutor. This tutor plays a crucial role, working closely with the students to refine the project, and providing essential validation and feedback from the external partner organization.

To maintain consistent progress, students are expected to regularly validate their work with the Academic Tutor, typically on a weekly basis. Additionally, they are required to present their proposals to the Partner Organization for approval before the final submission. It is the students' responsibility to proactively engage with their Academic Tutor and schedule mentoring sessions.

Their efforts will culminate in an evaluation by other professors and the Partner Organization during the final presentation, where they will showcase their analysis and recommendations.

Regular attendance in class sessions and active participation in group meetings are mandatory for all students. While the schedule for master classes is fixed for all groups, students are accountable for organizing their individual and collective work schedules and venues, as well as arranging their mentoring sessions.

Groups Formation

The companies will present the project and the students will express their interest, but the assignation will be performed by IE.

Project Development

To ensure academic quality in all the projects carried out by the students, they will be supervised by, on the one hand, an academic tutor and, on the other, a Sponsor from the company that has provided the data set. This has the purpose to carry out the continuous

evaluation of the work developed by the students in groups, as well as the contribution of each one of the students to the final deliverables.

The "academic tutor" will be a professional specialized in the subject, who will help students during the tutoring sessions to work in the process of solving the challenge proposed by the company.

Attendance to the tutoring sessions is mandatory and will have an impact in the individual grading of each student.

Project Presentation

The Final Presentation is made once the Impact Projects have been previously evaluated by the Tutors. Both the Academic Tutor and the Company Sponsor must approve that their groups have the minimum quality to present in the Final Presentation.

At the end of the time allotted for the completion of the Impact Project, all members of the work team must present the results obtained, as well as the proposed solution, to an evaluation panel. (**Expected date 13th July 2026**)

The final panel will be made up of tutors from the companies that provide the data sets and two professors. These members of the panel oversee the final evaluation of the work carried out, this being the deliverables and presentation.

The presentation is expected to be deliverer showing the communication skills and data visualization necessary for a professional work.

Learning Activity	Weighting
Group work	100.0 %
TOTAL	100.0 %

AI POLICY

#1 – Critical GenAI use is encouraged

In this course, **the use of generative artificial intelligence (GenAI) is encouraged**, with the goal of developing an informed critical perspective on potential uses and generated outputs.

However, be aware of the limits of GenAI in its current state of development:

· If you provide minimum effort prompts, you will get low quality results. You will need to refine your prompts to get good outcomes. This will take work.

· Don't take ChatGPT's or any GenAI's output at face value. Assume it is wrong unless you either know the answer or can cross-check it with another source. You are responsible for any errors or omissions. You will be able to validate the outputs of GenAI for topics you understand.

· AI is a tool, but one that you need to acknowledge using. Failure to do so is in violation of academic honesty policies. Acknowledging the use of AI will not impact your grade.

Suggested format to acknowledge the use of generative AI tools:

I acknowledge the use of [AI systems link] to [specify how you used generative AI]. The prompts used include [list of prompts]. The output of these prompts was used to [explain how you used the outputs in your work].

If you have chosen not to include any AI generated content in your assignment, the following disclosure is recommended:

No content generated by AI technologies has been used in this assignment.

PROGRAM

SESSION 1 (LIVE IN-PERSON)

Capstone Project Info Session. **31 January 2026**

We will the present the Capstone Project Goal and Objectives

SESSION 2 (LIVE ONLINE)

Proposed Challenges. 20 March 2025

We will introduce the participating companies and their proposed challenges. Students will have the opportunity to hear directly from the companies about their data challenges and ask any questions they may have.

SESSION 3 (DISCUSSION FORUM)

Weekly forum to address any questions about Project Proposals. Week, March 13th 2026

SESSION 4 (LIVE ONLINE)

Deep dive into the corporate project deliverables - 30th May 2026

SESSION 5 (DISCUSSION FORUM)

Questions about project deliverables. Week 1st of June 2026

SESSION 6 (LIVE ONLINE)

Final tips Presentations. 4th July 2026

SESSION 7 (DISCUSSION FORUM)

Submissions of deliverables. Week 6th of July

SESSION 8 (LIVE IN-PERSON)

Presentations, 13th July 2026

SESSIONS 9 - 10 (LIVE ONLINE)

Weekly working sessions

During the weekly tutoring sessions, the assigned professor will provide individual support to each student team by answering questions, giving guidance, monitoring progress, and providing

constructive feedback on the work completed up to that point.

Students will organise themselves with the tutor to find the best moment of the week to meet, but once decided it is **mandatory** to attend.

Lack of a minimum attendance of 80% may end up in a direct fail.

SESSIONS 11 - 12 (LIVE ONLINE)

Weekly working sessions

During the weekly tutoring sessions, the assigned professor will provide individual support to each student team by answering questions, giving guidance, monitoring progress, and providing

constructive feedback on the work completed up to that point.

Students will organise themselves with the tutor to find the best moment of the week to meet, but once decided it is **mandatory** to attend.

Lack of a minimum attendance of 80% may end up in a direct fail.

SESSIONS 13 - 14 (LIVE ONLINE)

Weekly working sessions

During the weekly tutoring sessions, the assigned professor will provide individual support to each student team by answering questions, giving guidance, monitoring progress, and providing

constructive feedback on the work completed up to that point.

Students will organise themselves with the tutor to find the best moment of the week to meet, but once decided it is **mandatory** to attend.

Lack of a minimum attendance of 80% may end up in a direct fail.

SESSIONS 15 - 16 (LIVE ONLINE)

Weekly working sessions

During the weekly tutoring sessions, the assigned professor will provide individual support to each student team by answering questions, giving guidance, monitoring progress, and providing

constructive feedback on the work completed up to that point.

Students will organise themselves with the tutor to find the best moment of the week to meet, but once decided it is **mandatory** to attend.

Lack of a minimum attendance of 80% may end up in a direct fail.

SESSIONS 17 - 18 (LIVE ONLINE)

Weekly working sessions

During the weekly tutoring sessions, the assigned professor will provide individual support to each student team by answering questions, giving guidance, monitoring progress, and providing

constructive feedback on the work completed up to that point.

Students will organise themselves with the tutor to find the best moment of the week to meet, but once decided it is **mandatory** to attend.

Lack of a minimum attendance of 80% may end up in a direct fail.

SESSIONS 19 - 20 (LIVE ONLINE)

Weekly working sessions

During the weekly tutoring sessions, the assigned professor will provide individual support to each student team by answering questions, giving guidance, monitoring progress, and providing

constructive feedback on the work completed up to that point.

Students will organise themselves with the tutor to find the best moment of the week to meet, but once decided it is **mandatory** to attend.

Lack of a minimum attendance of 80% may end up in a direct fail.

SESSIONS 21 - 22 (LIVE ONLINE)

Weekly working sessions

During the weekly tutoring sessions, the assigned professor will provide individual support to each student team by answering questions, giving guidance, monitoring progress, and providing

constructive feedback on the work completed up to that point.

Students will organise themselves with the tutor to find the best moment of the week to meet, but once decided it is **mandatory** to attend.

Lack of a minimum attendance of 80% may end up in a direct fail.

SESSIONS 23 - 24 (LIVE ONLINE)

Weekly working sessions

During the weekly tutoring sessions, the assigned professor will provide individual support to each student team by answering questions, giving guidance, monitoring progress, and providing

constructive feedback on the work completed up to that point.

Students will organise themselves with the tutor to find the best moment of the week to meet, but once decided it is **mandatory** to attend.

Lack of a minimum attendance of 80% may end up in a direct fail.

SESSIONS 25 - 26 (ASYNCHRONOUS)

Students work in the project

SESSIONS 27 - 28 (ASYNCHRONOUS)

Students work in the project

SESSIONS 29 - 30 (ASYNCHRONOUS)

Students work in the project

EVALUATION CRITERIA

REPORT AND PRESENTATION

The final report should provide a comprehensive overview of the proposed solution to the problem or challenge presented by the company with the following sections.

- **Objective**: this section should clearly define the problem being addressed and the target of the work to be carried out.

- **Data sources**: this section should detail the data used in the analysis, including any preparation or processing that was done, and any additional data that was needed and how it was obtained.

- **Methodology**: this section should describe the tools and processes used to solve the problem, as well as the technology used.

- **Potential solution**: this section should present at least two alternatives that were considered throughout the resolution process, with an explanation of why one was chosen over the others

- **Development**: this section should include the code used to solve the problem, adapted to established standards for ease of understanding by a specialist in the field.

- **Conclusions**: Finally, the conclusions section should present the result from a technical perspective, including any metrics such as algorithm choice and precision, as well as its potential impact and viability for implementation in the business.

The report, along with the information included in it, will be meticulously evaluated by both the academic tutor and the company sponsor throughout the tutoring sessions leading up to the final presentation before the panel. The report must receive the endorsement of both the company sponsor and academic tutor before the group can present their solution to the panel.

PRESENTATION GUIDELINES

One of the most important evaluation criteria will be professionalism, i.e. the capability to inspire confidence in the analysis and proposed strategy. Therefore, bear in mind the following points:

- **Professional setting**: You should not act as if you were before a panel of professors, but rather as a member of the team of business analytics presenting in front of executive management from the company.

- **Focus on analysis and recommendations**: You should not begin by listing the characteristics of the company or its sector, given that the audience will be more than aware of them already. Rather, we want you to analyze the specific problem or challenge, propose alternatives using the tools and knowledge you developed, and make data-driven decisions relevant to the company’s situation. The audience is there to hear the final conclusions of your analysis, but also to understand the way you reached it. Organize your presentation and report around the insights and recommendations you want to transmit (and the information and data backing them up), but do not forget to include the process and tools you followed to discover them.

- **Use tools wisely and avoid one size fits all:** You should not try to show that you are an expert in a vast number of models and tool, but rather that you can identify management problems and of providing feasible, realistic solutions using the technical tools you have acquired during the program.

- **Synthesize**: The presentation should be clear, specific, and concise. Do not waste time mentioning anything that has no direct bearing on an important conclusion. When presenting numerical data, just briefly point out the assumptions (unless they are obvious) behind said

numbers and the results that you consider important.

- **Design and content:** You should use slides, but you need to make sure you have the right quantity and quality of slides to keep the audience's attention.

- **Time management**: It is advisable to spend a certain amount of time before the exam preparing the structure of the presentation to ensure optimal time allocation. Remember, however, that although the exam usually goes as planned, the exam panel may suggest skipping a point or ask for more detail at any point.

DELIVERABLES

Each group must submit the following deliverables.

- **Executive Summary**: A short description of the problem and solution (in PDF format) to scitech@ie.edu and uploading to blackboard. Max. 2 pages (in PDF format). It can also be an A3/poster with the main aspects of the solution proposed.

- Written Report: (in PDF format) via e-mail scitech@ie.edu and uploading to blackboard on

the date the **program management will confirm, typically 3 to 5 workdays** before the final presentation date. The format must be Min. 10 pages, Max. 20 pages (not counting appendices), 1.0 spacing, 12- point Times New Roman font, with proper citation. File name should indicate the content and incorporate the name of the project and the team members' primary surnames.The cover page should include the team members' primary surnames.

- Final Presentation: send a pdf version of your presentation via email to sci-tech@ie.edu by 8:30 am on the day of your final presentation.

- **Code**: Github repository url, added to the executive summary.

- Ethics Statement: The cover page of the Written Report must contain the following statement and be accompanied by the students' signatures. Please ensure that each team member signs the cover page (digital signatures). "I hereby certify that this report and the accompanying presentation is my own original work in its entirety, unless where indicated and referenced."

- **Citations**: You need to properly cite and reference all your external sources in the report. It is not necessary to cite the source of the pictures you may be using in your presentation.

However, if you incorporate graphics or charts with information content into the presentation or the report, you should include a small note at the bottom of the slide or graphic indicating the source, or their originality when they are created by you.

- **Final Presentation**: There is no minimum or maximum number of slides required in the presentation, but students must observe the time allotment for the presentation.

File name should indicate: IMPACT PROJECT_PRESENTATION_name of the project.

ACADEMIC TUTOR ROLE

The role of the academic tutors is to help follow up, to evaluate the evolution of the group, and to make sure that the group is advancing properly in the project. It is a technical advisor, but will not participate in the analysis, nor the decision-making process.

The tutors duties do not include to solve specific doubts related to code bugs, detection of errors in settings, evaluation or validation of models, or similar ones.

Students have been prepared during the entire course to perform these tasks autonomously and errors they may make in the specific tasks of data preparation or modeling will not be considered

responsibility of the tutors.

The tutor will evaluate the members of the group individually, will control attendance to the agreed meetings and will provide the grade before the Final Presentation.

Finally, it is the tutor's prerogative to tell the students the final grade granted to each of the members of the group, before the Final Presentation.

There is also a Technical Coordinator, Prof. Federico Castanedo, who will provide to general sessions on how to address professionally the project.

RECOMMENDATIONS

- Business Understanding. Students should work on a technical proposal according to the challenge presented by the partner organization in the introductory session. The proposals must include the necessary elements that allow the company to evaluate each one of them (goals, teams, planned worked, durations, responsibilities, limitations, etc). The proposal will be considered as part of the evaluation.

- Data Understanding. Students should work on the understanding of the data before starting the analytical tasks themselves. These tasks include to determine what data is needed, to explore it, verify data quality, process and prepare it, visualize it. Some descriptive analysis is normally desired. Data audit process will be part of the evaluation.

- Data preparation. One of the most important tasks in a project that normally makes the difference in the outcome. However, is simultaneously the most time consuming and least rewarding one. Tasks include cleaning data, merging data sources, generate new attributes or transform existing

variables. Goal is to enrich datasets and prepare them for modeling. Data preparation will be considered as part of the course evaluation.

- Modeling & Evaluation. The students will be responsible for designing the modeling strategy and executing the technical tasks. Each group must evaluate and validate its analysis proposals as the basis for the analytical solution to be offered to company's managers. The design of the modelling strategy as well as the technical work done, and the quality of the results will be part of the evaluation.

criteria	percentage	Learning Objectives	Comments
Group Presentation	75 %		Panel Presentation
Other	25 %		Academic Tutor Grade

FAILING GRADE AND REASSESSMENT

When students receive a Fail in a course, they have the opportunity to present themselves for reassessment in order to earn the necessary credits toward graduation.

The reassessment of students should be scheduled between 5 and 10 working days after the review session takes place.

Grades for the reassessment are limited to a Low Pass and Fail.

Both, the initial Fail as well as the grade of the reassessment remain on the transcript. For the purpose of calculating the GPA however, only the grade of the reassessment is to be considered. Students receiving a failing grade in the reassessment of a course will not be able to continue in the program.

BIBLIOGRAPHY

Recommended

- Foster Provost; Tom Fawcett. (2013). *Data science for business : what you need to know about data mining and data-analytic thinking.* 1st. ISBN 9781449361327 (Digital)

BEHAVIOR RULES

Please, check the University's Academic Integrity Policy here. The Program Director may provide further indications.

ATTENDANCE POLICY

Please, check the University's Attendance Policy here. The Program Director may provide further indications.

ETHICAL POLICY

Please, check the University's Student Code of conduct here. The Program Director may provide further indications.